Residual collector with distinguished points and an equivariant walk (priced)

Correction. `research/ic_residual_dp_20261101` left the per-step distinguished-point test out of the priced counters, so its w > 0 figures for `residual-fused-dp` understate priced work. This study reruns with the per-step work priced (`*_uncharged` , 1 GAE per operation at w = 1, like rho's canonicalisation) and supersedes those figures.

Pricing caveat (from the independent validator, added after validation). In the band as built, `residual_table_bytes_uncharged` is a *byte* count (table entries × (16 + 8·columns)), but its `_uncharged` name makes the band price it as *operations* at weight w. For the baseline collector it is 86–92% of priced cost at w = 1 and 52–69% at w = 0.1; the candidates store only distinguished points, so their counter is about 150–210× smaller. **The large w > 0 candidate/baseline gains and the 48–53× baseline distances to rho in this report are therefore mostly an artifact of that pricing, not operation-count savings.** The w = 0 results are unaffected. The tables below put the band as built beside the band with that one counter excluded (same data, same pairing).

curve	ratio (w = 0 / 0.1 / 1)	as built	byte counter excluded
p36	res1 → res1eq	0.64 / 0.35 / 0.16	0.64 / 0.72 / 1.08
p36	res4 → res4eq	0.68 / 0.24 / 0.09	0.68 / 0.76 / 1.14
p36	res1 → res1dp	0.88 / 0.42 / 0.13	0.88 / 0.88 / 0.88
p36	res4 → res4dp	0.87 / 0.28 / 0.07	0.87 / 0.87 / 0.87
p36	rho → res4	3.97 / 12.62 / 51.57	3.97 / 3.97 / 3.97
p36	rho → res1eq	1.54 / 1.75 / 2.66	1.54 / 1.73 / 2.60
p36	rho → res1dp	2.13 / 2.14 / 2.22	2.13 / 2.12 / 2.12
p37	res1 → res1eq	0.66 / 0.36 / 0.16	0.66 / 0.74 / 1.12
p37	res4 → res4eq	0.66 / 0.24 / 0.09	0.66 / 0.74 / 1.12
p37	res1 → res1dp	0.82 / 0.39 / 0.12	0.82 / 0.82 / 0.82
p37	res4 → res4dp	0.90 / 0.28 / 0.07	0.90 / 0.90 / 0.89
p37	rho → res4	4.00 / 12.71 / 51.92	4.00 / 4.00 / 4.00
p37	rho → res1eq	1.64 / 1.86 / 2.83	1.64 / 1.85 / 2.79
p37	rho → res1dp	2.03 / 2.04 / 2.07	2.03 / 2.03 / 2.03

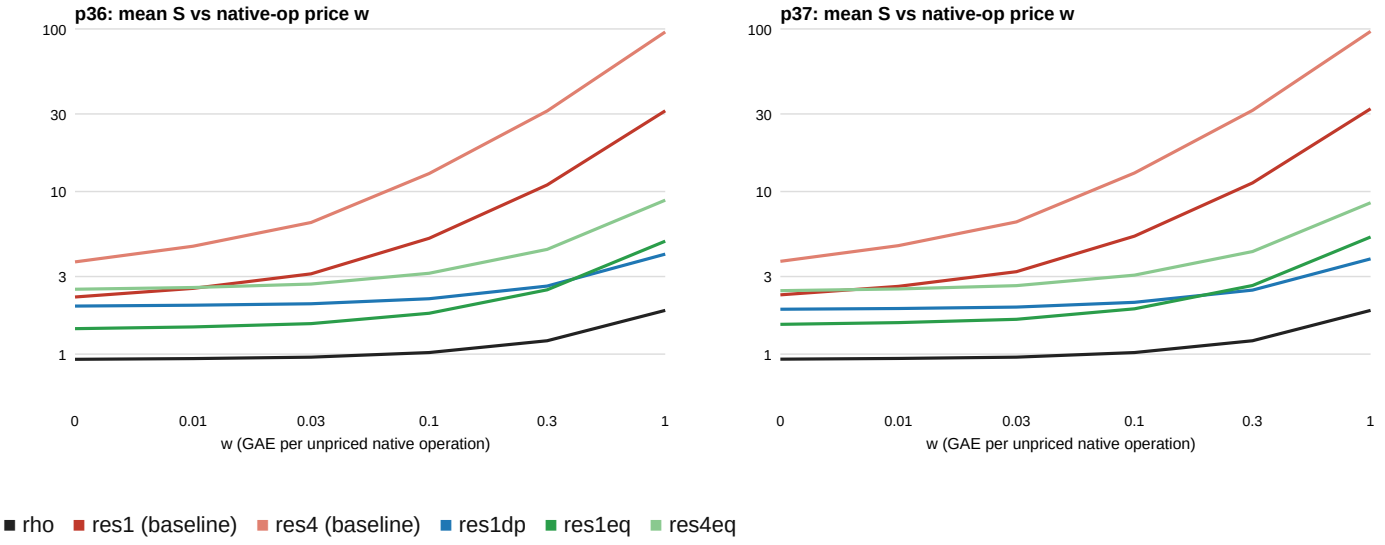
Reading with the byte counter excluded: **dp** is a constant 0.76–0.94× the baseline at every weight (a modest real saving; 0.44 for res4 on p42, whose baseline has one censored run); **eq** is 0.54–0.68× at w = 0 but 0.9–1.18× at w = 1 (0.55 for res4 on p42, same censored baseline) (it pays about 2.5 priced operations per step, so it is no better than the baseline at w = 1 once bytes are not counted). The distance to rho of the baseline collapses to 2.5–4.1× at every weight (res1 ≈ 2.5, res4 ≈ 3.4–4.1). Rho's own counters are also a lower bound (cycle-window scan, jump index and DP test not counted), which makes w > 0 distances to rho look larger than they are. Whether bytes should be priced as operations is the owner's call (§5a); nothing here changes the band.

What was implemented (`src/cryptanalysis/ic_framework/residual.rs` ; baseline `targets=residual` unchanged): `residual-fused-dp` (one addition per step from a $J_j - P_i$ table, distinguished points with $b = \lfloor \log_2 r / 2 \rfloor - 9 = 8$ on p36/p37) and `residual-fused-eq` (a negation-equivariant walk, so L and −L paths merge, on 2016 pair sums $J_u + J_v - P_i$ with a 2-cycle escape and distinguished points). Priced per step: DP test (both), orientation read and one negation per flipped step (eq). Setup additions and the extra escape additions are counted group operations.

Counted S ratio, candidate / baseline (paired by instance; 16 targets × 3 rounds; 95% interval; green = below 1)

curve	pair	w = 0	w = 0.1	w = 1
p36	res1dp/res1	0.880 [0.752, 1.030]	0.425 [0.363, 0.497]	0.132 [0.112, 0.154]
p36	res1eq/res1	0.639 [0.561, 0.727]	0.346 [0.304, 0.394]	0.158 [0.138, 0.180]
p36	res2dp/res2	0.838 [0.777, 0.904]	0.346 [0.321, 0.373]	0.099 [0.092, 0.107]
p36	res2eq/res2	0.654 [0.588, 0.728]	0.303 [0.272, 0.337]	0.127 [0.114, 0.141]
p36	res4dp/res4	0.871 [0.786, 0.964]	0.279 [0.252, 0.308]	0.073 [0.066, 0.081]
p36	res4eq/res4	0.679 [0.612, 0.752]	0.243 [0.219, 0.270]	0.092 [0.083, 0.102]
p37	res1dp/res1	0.816 [0.679, 0.980]	0.392 [0.327, 0.471]	0.119 [0.099, 0.143]
p37	res1eq/res1	0.660 [0.561, 0.776]	0.358 [0.304, 0.421]	0.163 [0.138, 0.192]
p37	res2dp/res2	0.941 [0.821, 1.079]	0.386 [0.337, 0.443]	0.108 [0.094, 0.124]
p37	res2eq/res2	0.671 [0.562, 0.802]	0.310 [0.259, 0.371]	0.129 [0.108, 0.155]
p37	res4dp/res4	0.895 [0.800, 1.002]	0.284 [0.254, 0.318]	0.072 [0.064, 0.081]
p37	res4eq/res4	0.660 [0.590, 0.737]	0.235 [0.211, 0.263]	0.088 [0.079, 0.099]

Frozen rule C1 (the 12 (curve, B, candidate) cells, below 1 at w = 0, 0.1, 1 in the band as built; the protocol text says "24", an arithmetic slip): **not met**. 9 of 12 cells hold: eq 6 of 6, dp 3 of 6; the three failures are all dp at w = 0 (p36 res1dp upper bound 1.030, p37 res2dp 1.079, p37 res4dp 1.002). A/A (res2 vs res2aa) = 1.0000 on both curves; 1408/1408 records verified; `ecbench verify --replay 12` audits OK.



Distance remaining to matched rho (S ratio rho → arm; 1.00 = parity)

curve	arm	w = 0	w = 0.1	w = 1
p36	res4	3.97	12.62	51.57
p36	res2	3.12	7.65	28.04
p36	res1	2.42	5.04	16.87
p36	res4dp	3.46	3.52	3.77
p36	res2dp	2.62	2.64	2.77
p36	res1dp	2.13	2.14	2.22

p36	res4eq	2.70	3.07	4.76
p36	res2eq	2.04	2.32	3.56
p36	res1eq	1.54	1.75	2.66
p37	res4	4.00	12.71	51.92
p37	res2	2.90	7.11	26.07
p37	res1	2.49	5.19	17.37
p37	res4dp	3.58	3.61	3.74
p37	res2dp	2.73	2.75	2.81
p37	res1dp	2.03	2.04	2.07
p37	res4eq	2.64	2.99	4.58
p37	res2eq	1.95	2.21	3.36
p37	res1eq	1.64	1.86	2.83

(Band as built.) Rho is ahead of every arm at every weight. The best arm at w = 0 is res1eq (1.54× p36, 1.64× p37 rho) and at w = 1 res1dp (2.22×, 2.07×); baseline res4 was 3.97 / 12.6 / 51.6× (p36) and 4.00 / 12.7 / 51.9× (p37).

eq versus dp, same B

curve	pair	w = 0	w = 0.1	w = 1	order
p36	res1eq/res1dp	0.726 [0.642, 0.820]	0.815 [0.721, 0.921]	1.201 [1.061, 1.359]	unresolved inside the band
p36	res2eq/res2dp	0.780 [0.699, 0.871]	0.876 [0.784, 0.978]	1.282 [1.145, 1.435]	unresolved inside the band
p36	res4eq/res4dp	0.780 [0.695, 0.875]	0.874 [0.778, 0.981]	1.262 [1.123, 1.418]	unresolved inside the band
p37	res1eq/res1dp	0.809 [0.699, 0.935]	0.911 [0.788, 1.054]	1.364 [1.177, 1.580]	unresolved inside the band
p37	res2eq/res2dp	0.713 [0.613, 0.830]	0.803 [0.690, 0.935]	1.196 [1.025, 1.395]	unresolved inside the band
p37	res4eq/res4dp	0.737 [0.669, 0.811]	0.829 [0.752, 0.913]	1.225 [1.111, 1.350]	unresolved inside the band

eq has the lower count at w = 0 and dp at w = 1 (eq pays about 2.5 priced operations per step against 1); the order is unresolved inside the band, so neither is ranked above the other.

Classification and caveats

- **Engineering** improvement of a generic collision search: the collector is a rho walk that needs B + 1 collisions, so its floor is about √(B + 1) times rho. At B = 1 the arms sit at 1.5–2.2× rho. It is not a structural index-calculus advance and does not beat rho.
- Per-step pricing is deliberately conservative: a DP test, an orientation read and negations are each charged 1 GAE at w = 1, where rho's tuned walk charges about 1 operation per step in total. A cheaper model (DP as a bit test on the canonical x, as in rho) would lower the eq w = 1 figures; it was not applied here.
- Only p36 and p37 (both ≈ 2^36) exercise the DP path; larger r is unmeasured. Local L0 run, counts only. No `VALIDATION.json`, so no leaderboard or `benchmarks/README.md` row.
- Exploratory, before this study: a first equivariant attempt with 16 jumps fell into short cycles (walks trapped until the cap) and lost; the 64-jump pair-sum step set with a 2-cycle escape fixed it.

Sources

Teske, Math. Comp. 70 (2001) 809–825; Bos–Kleinjung–Lenstra, ANTS-IX, LNCS 6197 (2010) 66–82 (fruitless cycles of the negation map; checked by search earlier in this session); van Oorschot–Wiener, J. Cryptology 12 (1999) 1–28 (distinguished points; cited from memory). Earlier work: `research/ic_residual_fused_20261031`, `research/ic_residual_dp_20261101`, `research/ic_prime_residual_lp_20261026`.